Michael Huang
EN.625.633.81
Homework 9

1.

Exercise 4.9 Show that

$$\int_{-\infty}^{\infty} \exp\{-x^2\} \exp\{-(x-\mu)^2 y/2\sigma^2\} \, \mathrm{d}x = \exp\left\{-\frac{\mu^2}{1+2\sigma^2/y}\right\}$$

by completing the square in the exponent to evaluate the integral.

We start with

$$\begin{aligned}
&\int_{-\infty}^{\infty} \exp\{-x^2\} \exp\{\frac{-(x-\mu)^2 y}{2\sigma^2}\} dx \\
&= \int_{-\infty}^{\infty} \exp\{-x^2 - \frac{(x-\mu)^2 y}{2\sigma^2}\} dx \\
&= \int_{-\infty}^{\infty} \exp\{-x^2 - \frac{(x^2+\mu^2-2x\mu)y}{2\sigma^2}\} dx \\
&= \int_{-\infty}^{\infty} \exp\{-x^2 - \frac{x^2 y}{2\sigma^2} - \frac{y\mu^2}{2\sigma^2} + \frac{xy\mu}{\sigma^2}\} dx \\
&= \int_{-\infty}^{\infty} \exp\{-(1+\frac{y}{2\sigma^2})x^2 + \frac{xy\mu}{\sigma^2} - \frac{y\mu^2}{2\sigma^2}\} dx
\end{aligned}$$

With this expression equating to the form of $-ax^2 + bx + c$, we can try to complete the square as instructed, which would yield $-a(x - \frac{b}{2a})^2 + \frac{b^2}{4a} + c$. Let's try to match the format with $a = 1 + \frac{y}{2\sigma^2}$, $b = \frac{y\mu}{\sigma^2}$, and $c = -\frac{y\mu^2}{2\sigma^2}$:

$$\begin{aligned}
\frac{b}{2a} &= \frac{\frac{y\mu}{\sigma^2}}{2 + \frac{y}{\sigma^2}} \\
&= \frac{\frac{y\mu}{\sigma^2}}{\frac{2\sigma^2+y}{\sigma^2}} \\
&= \frac{y\mu}{2\sigma^2 + y}
\end{aligned}$$

and

$$\frac{b^2}{4a} = \frac{\frac{y^2\mu^2}{\sigma^4}}{4 + \frac{2y}{\sigma^2}}$$

$$= \frac{\frac{y^2\mu^2}{\sigma^4}}{\frac{4\sigma^2+2y}{\sigma^2}}$$

$$= \frac{y^2\mu^2}{4\sigma^4+2y\sigma^2}$$

So plugging in:

$$\int_{-\infty}^{\infty} \exp\{-(1+\frac{y}{2\sigma^2})(x-\frac{y\mu}{2\sigma^2+y})^2 + \frac{y^2\mu^2}{4\sigma^4+2y\sigma^2} - \frac{y\mu^2}{2\sigma^2}\}dx$$

$$= \int_{-\infty}^{\infty} \exp\{-(1+\frac{y}{2\sigma^2})(x-\frac{y\mu}{2\sigma^2+y})^2 + \frac{y^2\mu^2 - y\mu^2(2\sigma^2+y)}{2\sigma^2(2\sigma^2+y)}\}dx$$

$$= \int_{-\infty}^{\infty} \exp\{-(1+\frac{y}{2\sigma^2})(x-\frac{y\mu}{2\sigma^2+y})^2 + \frac{y^2\mu^2 - 2y\mu^2\sigma^2 - y^2\mu^2}{2\sigma^2(2\sigma^2+y)}\}dx$$

$$= \int_{-\infty}^{\infty} \exp\{-(1+\frac{y}{2\sigma^2})(x-\frac{y\mu}{2\sigma^2+y})^2 + \frac{-2y\mu^2\sigma^2}{2\sigma^2(2\sigma^2+y)}\}dx$$

$$= \int_{-\infty}^{\infty} \exp\{-(1+\frac{y}{2\sigma^2})(x-\frac{y\mu}{2\sigma^2+y})^2 + \frac{-y\mu^2}{2\sigma^2+y}\}dx$$

$$= \exp\{\frac{-y\mu^2}{2\sigma^2+y}\} \int_{-\infty}^{\infty} \exp\{-(1+\frac{y}{2\sigma^2})(x-\frac{y\mu}{2\sigma^2+y})^2\}dx$$

The integral is the Gaussian integral, with the form $\int_{-\infty}^{\infty} e^{-ax^2}dx = \sqrt{\frac{\pi}{a}}$. Using this:

$$= \exp\{\frac{-y\mu^2}{2\sigma^2+y}\}\sqrt{\frac{\pi}{1+\frac{y}{2\sigma^2}}}$$

$$= \exp\{\frac{-y\mu^2}{2\sigma^2+y}\}\sqrt{\frac{\pi}{\frac{2\sigma^2+y}{2\sigma^2}}}$$

$$= \exp\{\frac{-y\mu^2}{2\sigma^2+y}\}\sqrt{\frac{2\pi\sigma^2}{2\sigma^2+y}}$$

With the additional term in the corrected version of the problem:

$$= \frac{1}{\sqrt{\frac{2\pi\sigma^2}{y}}} \exp\{\frac{-y\mu^2}{2\sigma^2+y}\}\sqrt{\frac{2\pi\sigma^2}{2\sigma^2+y}}$$

$$= \sqrt{\frac{y}{2\pi\sigma^2}} \exp\{\frac{-y\mu^2}{2\sigma^2+y}\}\sqrt{\frac{2\pi\sigma^2}{2\sigma^2+y}}$$

$$
\begin{aligned}
&= \sqrt{\frac{y}{2\pi\sigma^2} \cdot \frac{2\pi\sigma^2}{2\sigma^2 + y}} \exp\{\frac{-y\mu^2}{2\sigma^2 + y}\} \\
&= \sqrt{\frac{y}{2\sigma^2 + y}} \exp\{\frac{-y\mu^2}{2\sigma^2 + y}\} \\
&= \sqrt{\frac{y}{2\sigma^2 + y}} \exp\{\frac{-y\mu^2}{2\sigma^2 + y}\} \\
&= \frac{1}{\sqrt{\frac{2\sigma^2+y}{y}}} \exp\{\frac{-y\mu^2}{2\sigma^2 + y}\} \\
&= \frac{1}{\sqrt{\frac{2\sigma^2}{y} + 1}} \exp\{\frac{-y\mu^2}{2\sigma^2 + y}\} \\
&= \frac{1}{\sqrt{\frac{2\sigma^2}{y} + 1}} \exp\{\frac{-y\mu^2}{2\sigma^2 + y}\} \\
&= \frac{1}{\sqrt{\frac{2\sigma^2}{y} + 1}} \exp\{-\frac{\mu^2}{\frac{2\sigma^2+y}{y}}\} \\
&= \boxed{\frac{1}{\sqrt{\frac{2\sigma^2}{y} + 1}} \exp\{-\frac{\mu^2}{1 + \frac{2\sigma^2}{y}}\}}
\end{aligned}
$$

exactly as we sought to show in the corrected version of the problem.

2.

Exercise 4.10 Consider an Accept–Reject method for the target density f based on the instrumental distribution g.

a. Show that the Accept–Reject sample $(X_1, \ldots, X_n)$ can be associated with two iid samples, $(U_1, \ldots, U_N)$ and $(Y_1, \ldots, Y_N)$, with corresponding distributions $\mathcal{U}_{[0,1]}$ and g; N is the stopping time associated with the acceptance of n variables Y_j.
b. Deduce that the corresponding estimator of $\mathbb{E}_f[h(X)]$ based on $(X_1, \ldots, X_n)$ can be written as

$$\delta_1 = \frac{1}{n} \sum_{i=1}^{n} h(X_i) = \frac{1}{n} \sum_{j=1}^{N} h(Y_j)\, \mathbb{I}_{U_j \leq w_j}\,,$$

with $w_j = f(Y_j)/Mg(Y_j)$.

(a)

(b)

3.

Exercise 4.15 For each of the following cases, generate random variables X_i and Y_i and compare the empirical average and Rao–Blackwellized estimator of $\mathbb{E}_f(X)$ and $\mathrm{var}_f(X)$:

a. $X|y \sim \mathcal{P}(y)$, $Y \sim \mathcal{G}a(a, b)$ (X is negative binomial);
b. $X|y \sim \mathcal{N}(0, y)$, $Y \sim \mathcal{G}a(a, b)$ (X is a generalized t);
c. $X|y \sim \mathcal{B}in(y)$, $Y \sim \mathcal{B}e(a, b)$ (X is beta-binomial).

We aim to generate random variables X_i and Y_i and compare the empirical average and Rao-Blackwellized estimator of $E_f(X)$ and $\mathrm{Var}_f(X)$:

(b)

$X|y \sim N(0, y), Y \sim \mathrm{Gamma}(a, b)$, i.e. X is a generalized t:

```
> nmbSimulations = 1000
> n = 1000
> a = 5
> b = 1
>
> bruteForce_3b = function(nmbSimulations, n, a, b) {
+
+   # Need to keep mean and mean^2 for calculating variance
+   means = rep(0, nmbSimulations)
+   means_squared = rep(0, nmbSimulations)
+
+   for (i in 1:nmbSimulations) {
+
+     # Simulate n values of Y from Gamma(a,b)
+     y = rgamma(n, shape = a, scale = b)
+
+     # For each Y value, simulate X|Y ~ N(0, y)
+     x = rnorm(n, mean = 0, sd = sqrt(y))
+
+     means[i] = mean(x)
+     means_squared[i] = mean(x^2)
+   }
+
+   variance = mean(means_squared) - (mean(means))^2
+
+   print(paste("Empirical results with",
+               "nmbSimulations =", nmbSimulations,
+               ", n =", n,
+               ", a =", a,
+               ", b =", b
```

```
+               ))
+   print(paste("mean =", mean(means), "and variance =", variance))
+ }
>
> rao_blackwell_3b = function(nmbSimulations, n, a, b) {
+
+   # Need to keep mean and mean^2 for calculating variance
+   means = rep(0, nmbSimulations)
+   means_squared = rep(0, nmbSimulations)
+
+   for (i in 1:nmbSimulations) {
+
+     # Simulate n values of Y from Gamma(a,b)
+     y = rgamma(n, shape = a, scale = b)
+
+     # Calculate conditional expectation E(X|Y)
+     # Since X|Y ~ N(0, y), we have E(X|Y) = 0
+     condExpectation = rep(0, n)
+
+     means[i] = mean(condExpectation)
+     means_squared[i] = mean(condExpectation^2)
+   }
+
+   variance = mean(means_squared) - (mean(means))^2
+
+   print(paste("Rao-Blackwellized results with",
+               "nmbSimulations =", nmbSimulations,
+               ", n =", n,
+               ", a =", a,
+               ", b =", b
```

```
+   ))
+   print(paste("mean =", mean(means), "and variance =", variance))
+ }
>
> bruteForce_3b(nmbSimulations, n, a, b)
[1] "Empirical results with nmbSimulations = 1000 , n = 1000 , a = 5 ,
b = 1"
[1] "mean = 0.00101562111045003 and variance = 5.01758577452203"
> rao_blackwell_3b(nmbSimulations, n, a, b)
[1] "Rao-Blackwellized results with nmbSimulations = 1000 , n = 1000 ,
a = 5 , b = 1"
[1] "mean = 0 and variance = 0"
```

We see that the variance is set to 0 in the Rao-Blackwellized estimator. The variability in the variable generation was completely removed by using the conditional expectation instead of actually simulating X values from the normal distribution, which introduces actual variance.

(c)

$X|y \sim Bin(y), Y \sim \text{Beta}(a, b)$, i.e. X is a beta-binomial t:

```
asdf> nmbSimulations = 1000
> n = 1000
> m = 10
> a = 2
> b = 2
>
> bruteForce_3c = function(nmbSimulations, n, m, a, b) {
+
+   # Need to keep mean and mean^2 for calculating variance
+   means = rep(0, nmbSimulations)
+   means_squared = rep(0, nmbSimulations)
```

```
+
+   for (i in 1:nmbSimulations) {
+
+     # Simulate n values of Y from Beta(a,b)
+     y = rbeta(n, shape1 = a, shape2 = b);
+
+     # For each Y value, simulate X|Y ~ Binomial(m, y)
+     x = rbinom(n, size = m, prob = y);
+
+     means[i] = mean(x)
+     means_squared[i] = mean(x^2)
+   }
+
+   variance = mean(means_squared) - (mean(means))^2
+
+   print(paste("Empirical results with",
+               "nmbSimulations =", nmbSimulations,
+               ", n =", n,
+               ", m =", m,
+               ", a =", a,
+               ", b =", b
+   ))
+   print(paste("mean =", mean(means), "and variance =", variance))
+ }
>
> rao_blackwell_3c = function(nmbSimulations, n, m, a, b) {
+
+   # Need to keep mean and mean^2 for calculating variance
+   means = rep(0, nmbSimulations)
+   means_squared = rep(0, nmbSimulations)
```

```
+
+   for (i in 1:nmbSimulations) {
+
+     # Simulate n values of Y from Beta(a,b)
+     y = rbeta(n, shape1 = a, shape2 = b);
+
+     # Calculate conditional expectation E(X|Y)
+     # Since X|Y ~ Binomial(m, y), we have E(X|Y) = m * y
+     condExpectation = m * y;
+
+     means[i] = mean(condExpectation)
+     means_squared[i] = mean(condExpectation^2)
+   }
+
+   variance = mean(means_squared) - (mean(means))^2
+
+   print(paste("Rao-Blackwellized results with",
+               "nmbSimulations =", nmbSimulations,
+               ", n =", n,
+               ", m =", m,
+               ", a =", a,
+               ", b =", b
+   ))
+   print(paste("mean =", mean(means), "and variance =", variance))
+ }
>
> bruteForce_3c(nmbSimulations, n, m, a, b)
[1] "Empirical results with nmbSimulations = 1000 , n = 1000 , m = 10 ,
a = 2 , b = 2"
[1] "mean = 5.001012 and variance = 6.998882975856"
```

```
> rao_blackwell_3c(nmbSimulations, n, m, a, b)
[1] "Rao-Blackwellized results with nmbSimulations = 1000 , n = 1000 ,
m = 10 , a = 2 , b = 2"
[1] "mean = 5.00453563874706 and variance = 4.99710120693035"
```

We see that the variance is reduced by a significant amount ($\frac{2}{7} =$ 28% or so).

4.

Exercise 4.18 A naïve way to implement the antithetic variable scheme is to use both U and $(1-U)$ in an inversion simulation. Examine empirically whether this method leads to variance reduction for the distributions (i) $f_1(x) = 1/\pi(1+x^2)$, (ii) $f_2(x) = \frac{1}{2}e^{-|x|}$, (iii) $f_3(x) = e^{-x}\mathbb{I}_{x>0}$, (iv) $f_4(x) = \frac{2}{\pi\sqrt{3}}\left(1+x^2/3\right)^{-2}$, and (v) $f_5(x) = 2x^{-3}\mathbb{I}_{x>1}$. Examine variance reductions of the mean, second moment, median, and 75th percentile.

To calculate the weights for the Rao–Blackwellized estimator of Section 4.6, it is necessary to derive properties of the distribution of the random variables in the Accept–Reject algorithm. The following problem is a rather straightforward exercise in distribution theory and is only made complicated by the stopping rule of the Accept–Reject algorithm.

(i)

(ii)

(iv)

5.

Assuming that we use antithetic variables to estimate some parameter of the standard normal distribution, we must prove that the covariance between X_i and Y_i is -1, i.e. $\text{Cov}(X_i, Y_i) = -1$.

We set up with $U \sim \text{Unif}(0,1)$. Using the hint with the normal distribution being symmetric about 0, we know that $\Phi(z) = 1 - \Phi(-z)$ for the range of U, and equivalently that

$$x = \Phi(z) = 1 - \Phi(-z)$$

$$\Phi(-z) = 1 - x$$

$$-z = \Phi^{-1}(1 - x)$$

$$-\Phi^{-1}(-x) = \Phi^{-1}(1 - x)$$

So taking advantage of this, let's set $X = \Phi^{-1}(U)$ and $Y = \Phi^{-1}(1 - U)$ accordingly. We can then determine that

$$Y = \Phi^{-1}(1 - U) = -\Phi^{-1}(-U) = -X$$

So therefore

$$\begin{aligned} \mathrm{Cov}(X_i, Y_i) &= \mathrm{Cov}(X_i, -X_i) = -\mathrm{Cov}(X_i, X_i) \\ &= -\mathrm{Var}(X_i) = \boxed{-1} \end{aligned}$$

just as we aimed to prove.